

Homework #3

Wednesday, April 16, 2025 4:14 PM

Layne
Woodruff

#19

Info given:

- (STUDENT INFO) sName, sSSN, sNum, sCAddress, sCPhone, sPAddress, sPPhone, sBdate, sSex, sClass, sMajorCode, sMinorCode, sDegreeProgram
- (DEPT INFO) dName, dCode, dOffice, dPhone, dCollege
- (COURSE INFO) cName, cDesc, cNum, cCredit, cLevel, cDept
- (SECTION INFO) secInstructorName, secSemester, secYear, secCourse, secNum
- (GRADES) sSSN, secNum, gGrade (+ section info since it refers to a section): secCourse, secSemester, secYear, secNum

Functional Dependencies:

- sSSN (student SSN)
 - sName, sCAddress, sCPhone, sPAddress, sPPhone, sBdate, sSex, sClass, sMajorCode, sMinorCode, sDegreeProgram (Possibly sNum, but assuming not for exchange students and sSSN would be null?)
- sNum (Student Number)
 - sName, sCAddress, sCPhone, sPAddress, sPPhone, sBdate, sSex, sClass, sMajorCode, sMinorCode, sDegreeProgram, sSSN (same question for exchange students but I think you would be able to pull the the value is null)
- dName (Department Name)
 - dCode, dOffice, dPhone, dCollege
- dCode (Department Code)
 - dName, dOffice, dPhone, dCollege
- cNum (Course Number)
 - cName, cDesc, cCredit, cLevel, cDept
- secCourse + secSemester + secYear + secNum
 - secInstructorName
- secCourse, secSemester, secYear, secNum, sSSN
 - gGrade (need the extra info since secNum isn't unique, only unique to a given time period)

Foreign Keys:

- Department.dCode
 - Student.sMajorCode, Student.sMinorCode, Course.cDept
- Student.sSSN
 - Grades.sSSN
- Course.cNum
 - Section.secCourse
- Section(secCourse, secSemester, secYear, secNum)
 - Grades(secCourse, secSemester, secYear, secNum)

Student

<u>sNum</u>	sSSN	sCAddress	sCPhone	sPAddress	sPPhone	sBdate	sSex	sClass	sMajorCode	sMinorCode	sDegreeProgram
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Department

<u>dCode</u>	dName	dOffice	dPhone	dCollege
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Course

<u>cNum</u>	cName	cDesc	cCredit	cLevel	cDept
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Section

<u>secNum</u>	<u>secCourse</u>	<u>secSemester</u>	<u>secYear</u>	secInstructorName
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Grades

<u>secNum</u>	<u>sNum</u>	<u>secSemester</u>	<u>secYear</u>	<u>secCourse</u>	gGrade
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#24

Given:

- R = {A, B, C, D, E, F, G, H, I}
- Functional Dependencies:

- $A+B \rightarrow C$
- $A \rightarrow D+E$
- $B \rightarrow F$
- $F \rightarrow G+H$
- $D \rightarrow I+J$
- The key is A,B (you can get everything from A+B).
- (A gives D,E,I,J)
- (B gives F, G, H)
- (you can only get C from A and B)

2NF

Table 1

A	D	E	I	J
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Table2

B	F	G	H
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Table3

A	B	C
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- Can split up Table 1 and Table 2 (due to indirect dependencies)

3NF

Table 1

A	D	E
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Table 2

D	I	J
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Table 3

B	F
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Table 4

F	G	H
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Table 5

A	B	C
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